

Akshay R K

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LinkedIn , GitHub

Professional Summary

Backend Engineer with six years of experience building scalable production systems and AI powered features. Proven track record delivering event driven streaming systems and backend APIs using Node.js, TypeScript, NestJS, Kafka, PostgreSQL, and Redis. Experienced in schema design and ORM frameworks such as Prisma and TypeORM, building GraphQL and REST APIs, implementing unit and integration tests with Jest, and establishing CI/CD pipelines with GitHub Actions, Docker, and Kubernetes. Strong product orientation with ability to collaborate across frontend, design, and data teams. Brings consistent AI and ML delivery experience integrating LLM APIs and building retrieval augmented features where relevant.

Core Technical Skills

- Backend: Node.js, TypeScript, NestJS, Express, FastAPI, Django
- Messaging: Kafka, Kinesis, event driven systems
- APIs: GraphQL, REST, tRPC
- Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Prisma ORM, TypeORM, schema design, indexing, performance tuning
- Testing: Jest, Mocha, pytest, unit testing, integration testing, end to end testing
- Cloud and DevOps: AWS CDK, Lambda, API Gateway, RDS, S3, EKS, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, monitoring
- AI and ML: OpenAI, Anthropic, LLM API integration, vector database pipelines
- Tools: Git, Jira, Agile, Scrum, Figma

Professional Experience

Lead Software Engineer, Akkodis India Pvt Ltd (Client Stellantis)

Nov 2023 – Present

Bangalore, India

- Built backend services in Node.js, TypeScript, and NestJS delivering automotive and supply chain features with strong focus on scalability and reliability.
- Designed and operated Kafka based event driven pipelines handling telemetry and business events with high throughput and low latency.
- Developed GraphQL schemas and REST APIs ensuring type safety and contract alignment across services.
- Designed PostgreSQL schema with Prisma ORM and indexing strategies for low latency access to high volume datasets.
- Applied Redis caching for hot data queries, reducing API latency significantly.
- Integrated LLM APIs (OpenAI, Anthropic) to deliver AI features for user assistance and analytics.
- Implemented automated CI/CD using GitHub Actions, AWS CDK, and containerization improving release cycle efficiency by 30 percent.
- Established comprehensive test coverage with Jest for unit and integration tests.

Lead Software Developer, BLP Industry.AI

Sep 2020 – Oct 2023

Bangalore, India

- Delivered Conserve.AI energy management platform in cloud and on premise deployments using Node.js, Python, Django, PostgreSQL, and MongoDB.
- Designed and optimized REST and GraphQL APIs for scalable enterprise integrations.
- Implemented Redis caching and SQL tuning improving throughput by 35 percent.
- Built modular backend services with TypeScript and Node.js following clean architecture and secure design principles.
- Led CI/CD adoption with GitHub Actions, Docker, and Kubernetes reducing QA effort and accelerating deployments.
- Worked closely with product managers and design teams to refine architecture and improve user experience.

Senior Developer, Montaigne Smart Business Solutions

Mar 2019 – Sep 2020

Bangalore, India

- Built backend architecture for Leadmirror AI marketing platform with Node.js, Python, Django, and PostgreSQL.
- Integrated machine learning features for SEO optimization, lead generation, and keyword scraping.
- Delivered frontend dashboards in React for campaign analytics and reporting.
- Implemented REST APIs and ensured maintainability through test automation and best practices.

Selected Projects

SquadBrain Multi Tenant Knowledge Hub

Current

- Built multi tenant knowledge platform integrating GitHub and Jira content with real time RFC editing using CRDTs (yjs).
- Implemented backend in Node.js, TypeScript, and NestJS with PostgreSQL, Prisma, and Redis deployed on Kubernetes with OpenTelemetry and OAuth SSO.

- Integrated evaluation harness for retrieval quality metrics including precision at k and hallucination rates.

ADA Connected Vehicle Platform

Current

- Developed ADA Web (React, TypeScript), ADA Server (Python, AWS CDK, PostgreSQL, Redis), and ADA Client for telemetry.
- Built Kafka based ingestion pipelines and REST and GraphQL APIs supporting real time vehicle monitoring.
- Deployed with AWS Lambda, API Gateway, Kinesis, RDS, and containerized services ensuring cost efficiency and uptime.

Conserve.AI Energy Management System

Production

- Delivered hybrid cloud and on premise EMS with predictive analytics and dashboards built with Django, PostgreSQL, Redis, AWS, and GCP.
- Optimized large scale energy data processing for high performance and compliance.

Education

B.E. Electrical and Electronics Engineering	RNS Institute of Technology, Bangalore	2015 – 2018
Diploma in Electrical and Electronics Engineering	MEI Polytechnic, Bangalore	2012 – 2015

Additional Information

- Languages: English (fluent), German (beginner)
- Work style: Agile, Scrum, metrics driven delivery, early stage product development